



East West University

*Department of Computer Science and Engineering (CSE) Semester:
(Fall, Year: 2025), B.Sc. in CS E*

Designing an Interconnected Multi-Building Network

Mini Project

Course Code: CSE405

Course Title: Computer Networking

Section: 5

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Submitted to:

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Submitted By:

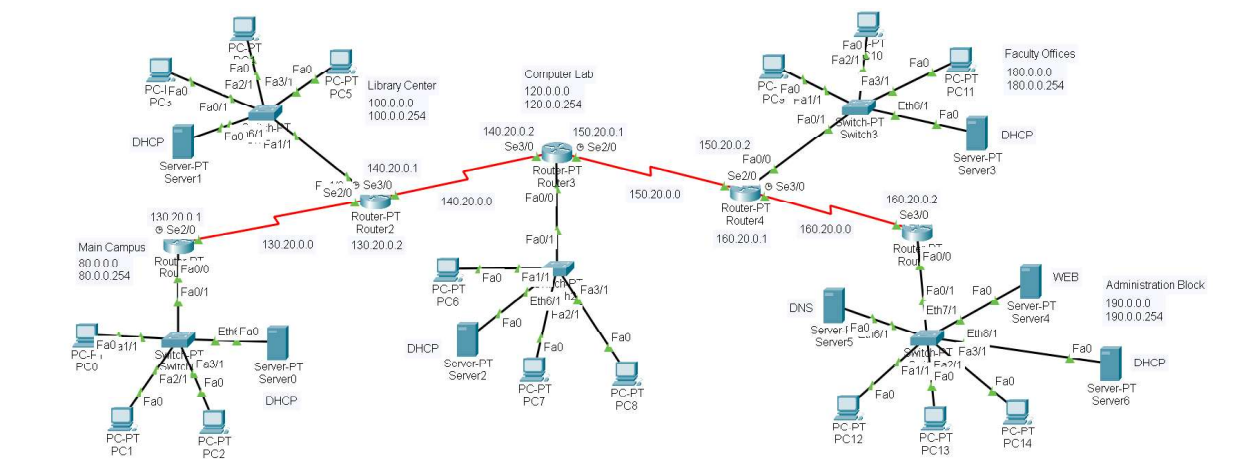
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Project Topic : Designing an Interconnected Multi-Building Network.

Problem Statement :

This project involves designing a five-facility campus network using a multi-router setup and EIGRP dynamic routing. It utilizes Class A and B hierarchical addressing for efficient segmentation, with dedicated DHCP servers in each building for automated IP management. A centralized DNS and Web server in the Administration Block provides institutional access, verified by successful cross-facility connectivity testing.

Network Design :



The network features five routers interconnected via serial links to form a core backbone. Each router connects to a local switch via FastEthernet, providing access to three PCs and a DHCP server per facility. This layered design separates LAN and WAN traffic, ensuring efficient communication and streamlined network management.

Router IP Configuration :

These commands assign IP addresses, enable interfaces, and establish serial communication between routers. The configuration is then saved to ensure all settings remain permanent.

```
Router>Enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 80.0.0.0 255.0.0.0
Bad mask /8 for address 80.0.0.0
Router(config-if)#ip address 80.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

These CLI commands assign IP addresses to the FastEthernet and Serial interfaces, activate them, and save the settings. The same process is repeated for all routers in the network.

EIGRP Protocol Configuration :

The EIGRP routing protocol is configured to enable dynamic routing between all routers in the network.

Router 1 (Main Campus) EIGRP config :

```
Router>Enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp
% Incomplete command.
Router(config)#router eigrp 1
Router(config-router)#network 80.0.0.0 0.255.255.255
Router(config-router)#network 130.20.0.0 0.0.255.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 2 (Library Center) EIGRP config :

```
Router>Enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 130.20.0.0 0.0.255.255
Router(config-router)#network 100.0.0.0 0.255.255.255
Router(config-router)#network 140.20.0.0 0.0.255.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 3 (Computer Lab) EIGRP config :

```
Router>Enable
Router#config t
Enter configuration commands, one per line.  End with
CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 120.0.0.0 0.0.255.255
Router(config-router)#network 140.20.0.0 0.0.255.255
Router(config-router)#network 150.20.0.0 0.0.255.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 4 (Faculty Offices) EIGRP config :

```
Router>Enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 150.20.0.0 0.0.255.255
Router(config-router)#network 160.20.0.0 0.0.255.255
Router(config-router)#network 180.0.0.0 0.0.255.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 5 (Administration Block) EIGRP config :

```
Router>Enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 160.20.0.0 0.0.255.255
Router(config-router)#network 190.0.0.0 0.0.255.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Network Service Integration :

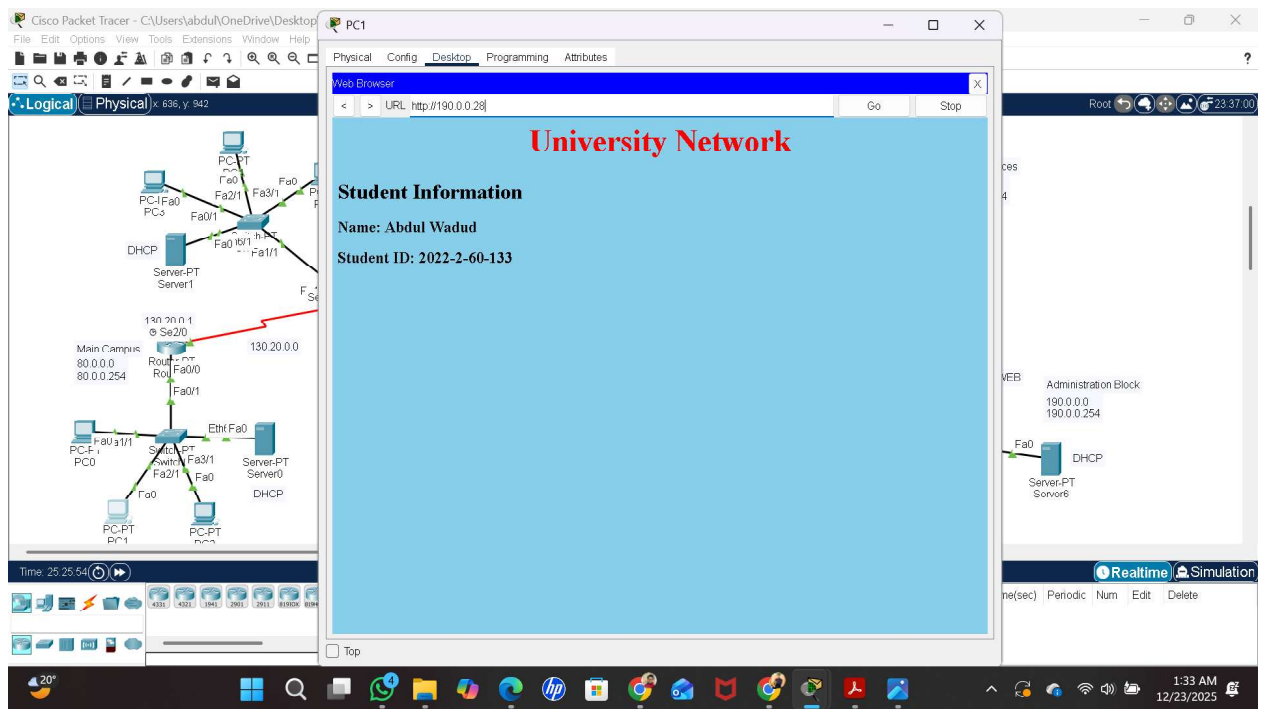
Each facility uses a dedicated DHCP server to automate IP, subnet, and DNS assignments for PCs, centralizing management and minimizing errors. The Administration Block hosts the DNS and Web Server for <http://www.universitynetwork.com>, which displays the student's name and ID, ensuring these services are accessible campus-wide.

Experimental Results :

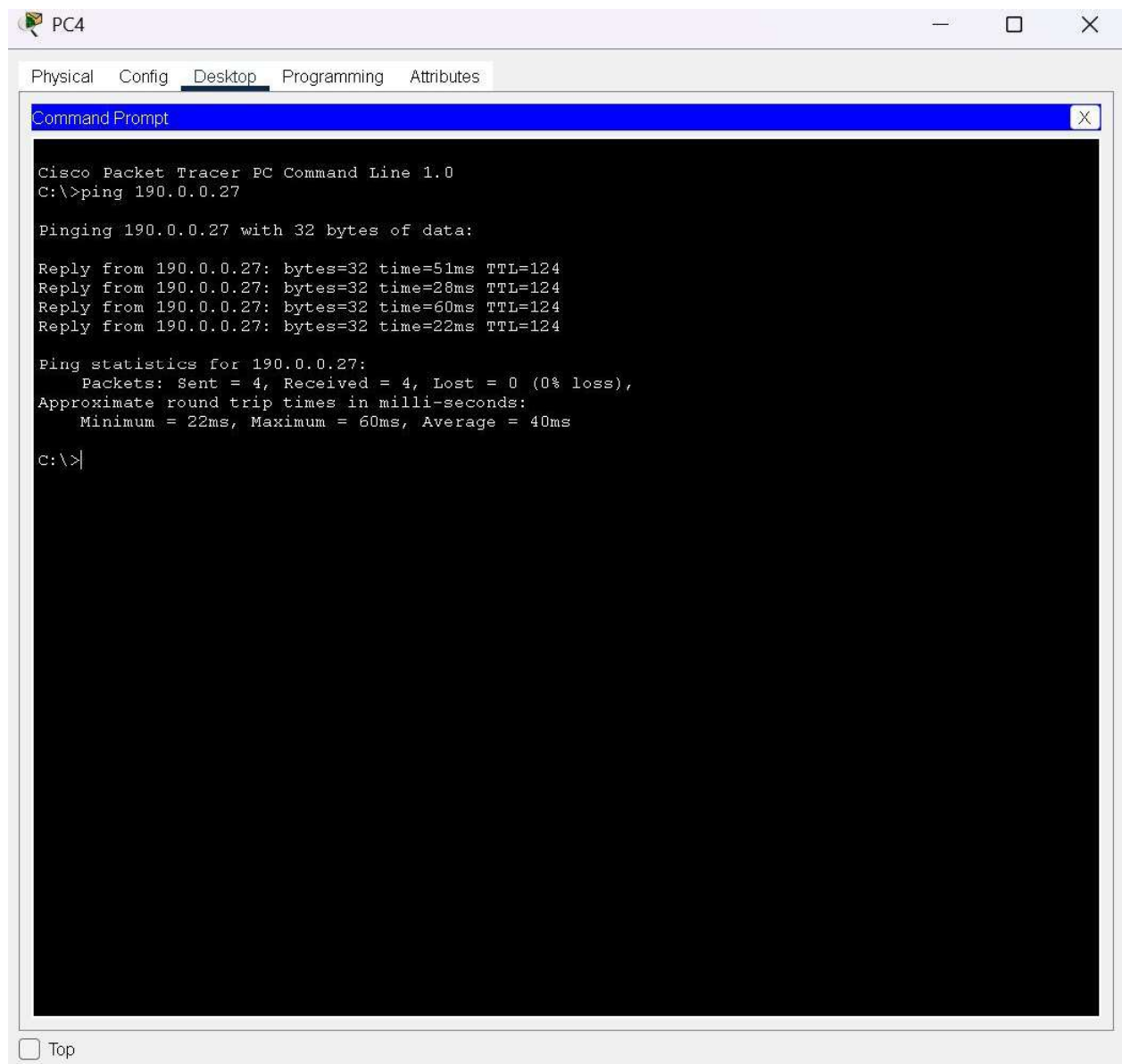
The implementation was verified through a series of systematic tests to ensure the network meets all design requirements. The experimental results are summarized below:

1. **Web Service Accessibility :** The Administration Block's centralized DNS and Web Servers are fully operational. All campus PCs successfully resolve <http://www.universitynetwork.com>, displaying the student's name and ID. This remote access confirms that EIGRP routing and DNS configurations are correctly

implemented across the entire network.



2. **Connectivity Validation** : A successful ping was performed between the 2nd PC of the Library Center and the 3rd PC of the Administration Block, confirming seamless inter-facility communication.



The image shows a screenshot of a Cisco Packet Tracer PC Command Line window for a device named PC4. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt shows the execution of a ping command to 190.0.0.27. The output indicates that the ping was successful, with 4 packets sent and received, and a 0% loss. The round trip times are listed as Minimum = 22ms, Maximum = 60ms, and Average = 40ms. The window also has a 'Top' button at the bottom left.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 190.0.0.27

Pinging 190.0.0.27 with 32 bytes of data:

Reply from 190.0.0.27: bytes=32 time=51ms TTL=124
Reply from 190.0.0.27: bytes=32 time=28ms TTL=124
Reply from 190.0.0.27: bytes=32 time=60ms TTL=124
Reply from 190.0.0.27: bytes=32 time=22ms TTL=124

Ping statistics for 190.0.0.27:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 60ms, Average = 40ms

C:\>
```

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Conclusion :

This project demonstrates a scalable campus network utilizing Class A and Class B addressing with EIGRP routing and DHCP automation. Centralized DNS and Web services in the Administration Block remain accessible across all segments, as verified by successful connectivity tests. The reliable design ensures efficient data transmission, making it an ideal infrastructure for modern educational institutions.